

FlowSim Pro

User Manual — Steady-State Multiphase Flow Simulator

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FlowSim Pro is an engineering application for steady-state multiphase flow in wells, tubing, flowlines, pipelines, and simple production networks. This manual describes how to start, navigate, and use the application in Cursor Cloud or on a local machine.

Table of Contents

- 1. Getting Started
- 2. Application Layout
- 3. Dashboard
- 4. Case Editor
- 5. Input Tabs Reference
- 6. Running a Simulation
- 7. Understanding Results
- 8. Nodal Analysis
- 9. Case Comparison & Sensitivity
- 10. Exporting Reports
- 11. Model Assumptions
- 12. Troubleshooting
- 13. API Reference (Optional)

1. Getting Started

1.1 Cursor Cloud (Recommended)

When using FlowSim Pro inside a Cursor Cloud workspace, the project files are already available. You do not need to download anything to your local PC.

Start the application with two terminals:

Terminal 1 – Backend (API & simulation engine):

```
cd flowsim-pro/backend  
pip install -r requirements.txt  
PYTHONPATH=. python3 -m uvicorn app.main:app --reload --port 8000
```

Terminal 2 – Frontend (web interface):

```
cd flowsim-pro/frontend  
npm install  
npm run dev
```

Open the app in your browser via Cursor port forwarding: **<http://localhost:3000>**

1.2 Local PC

To run on your own computer, clone the repository from GitHub and follow the same terminal commands above. Requirements: Python 3.10+, Node.js 18+.

2. Application Layout

The top navigation bar provides access to three main pages and a theme toggle:

Page	Purpose
Dashboard	Overview of all cases, quick solve, multi-case selection
Case Editor	Main workspace for editing inputs and viewing live results
Results	Full-size charts and case comparison viewer

Case Editor — Three-Panel Layout

Panel	Location	Function
Case Tree	Left	Create, select, duplicate, delete, and solve cases
Input Forms	Center-left	Edit fluid, well, flowline, boundary, nodal, heat inputs
Charts & Assumptions	Center	Pressure/temperature profiles, nodal curves, model info
Results & Diagnostics	Right	Summary numbers, warnings, solver messages

3. Dashboard

- View all saved simulation cases in a table.
- See summary metrics: liquid rate, bottomhole pressure (BHP).
- Click a case name to open it in the Case Editor.
- Click the Play (▶) icon to run a simulation without opening the editor.
- Click **+ New Case** to create a blank case.
- Check boxes on two or more cases, then click **Compare** to open the Results viewer.

On first startup, a sample case **Deviated Well - Alpha-1** is automatically loaded.

4. Case Editor

Top Toolbar

- **Case name** — editable text field; rename your case here.
- **Save** — saves current inputs to the database.
- **Solve** — runs the simulation and updates charts and results.
- **PDF / JSON / CSV** — export the case report in the selected format.

Case Tree (Left Panel)

- **+ New** — creates a new case with default inputs.
- Click a case to load it.
- Hover over a case for actions: Solve, Duplicate, Delete.

5. Input Tabs Reference

Fluid Tab

Parameter	Description / Units
Oil API	Oil API gravity (°API)
Gas Gravity	Gas specific gravity (air = 1)
GOR	Gas-oil ratio (scf/stb)
Water Cut	Fraction of water in total liquid (0–1)
Bubble Point	Bubble-point pressure (psi)
Reservoir Temp	Reservoir temperature (°F)
Water Salinity	Formation water salinity (ppm)

Well Tab

Parameter	Description / Units
Packer Depth	Packer setting depth MD (ft)
Perforation Depth	Perforation depth MD (ft)
Choke (64ths in)	Surface choke size in 64ths of an inch
Tubing Segments	MD/TVD top and bottom, inner diameter (in), inclination (°)

Flowline Tab

Parameter	Description / Units
Length	Segment length (ft)
ID	Inner diameter (in)
Elev Δ	Elevation change, positive = uphill (ft)
Ambient Temp	Surrounding temperature (°F)

Boundary, Nodal & Heat Transfer Tabs

Tab	Key Parameters
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Boundary	Liquid rate (stb/d), WHP (psi), optional BHP (psi)
Nodal	Reservoir pressure (psi), PI (stb/d/psi), IPR model (PI or Vogel)
Heat Transfer	Ambient temp (°F), U-value (Btu/hr·ft ² ·°F), burial, insulation

6. Running a Simulation

Recommended workflow:

- Select or create a case in the Case Tree.
- Enter fluid properties on the Fluid tab.
- Define tubing segments on the Well tab (use + Add for multiple segments).
- Add flowline segments if modeling surface lines.
- Set boundary conditions: rate and wellhead pressure.
- Configure nodal analysis parameters if studying inflow/outflow balance.
- Click **Save**, then click **Solve**.
- Review charts in the center panel and summary values on the right.

Boundary condition modes: If BHP is left blank, the solver marches top-down from WHP to calculate BHP (VLP mode). If BHP is specified, the solver marches bottom-up from BHP to calculate WHP.

7. Understanding Results

Results Panel (Right)

Output	Meaning
Liquid Rate	Total liquid production rate (stb/d)
WHP	Wellhead pressure (psi)
BHP	Bottomhole pressure at perforations (psi)
Flowline ΔP	Pressure drop across flowline segments (psi)
Nodal Rate	Operating rate from IPR/VLP intersection (stb/d)

Charts (Center)

- **Well Pressure & Temperature vs MD** — blue = pressure, red = temperature along measured depth.
- **Flowline Pressure Profile** — pressure vs distance along the pipeline.
- **Nodal Analysis (IPR/VLP)** — blue = inflow (IPR), orange = outflow (VLP), red dot = operating point.

8. Nodal Analysis

Nodal analysis finds the operating point where the Inflow Performance Relationship (IPR) intersects the Vertical Lift Performance (VLP) curve.

IPR Models

- **Linear PI** — $BHP = P_r - Q/PI$. Simple, good for undersaturated or moderate drawdown.
- **Vogel** — Non-linear IPR for solution-gas drive reservoirs below bubble point.

The operating point rate and BHP are shown in the Results panel and marked on the nodal chart. Convergence error indicates how closely IPR and VLP pressures match at the solution rate.

9. Case Comparison & Sensitivity

- On the Dashboard, select 2+ cases with checkboxes and click Compare.
- The Results page shows a bar chart and table comparing rate, WHP, BHP, and flowline ΔP .
- Sensitivity analysis is available via the API (</api/sensitivity>) for choke, WHP, tubing ID, GOR, and water cut.

10. Exporting Reports

Format	Contents
PDF	Engineering report with assumptions, summary table, fluid properties, warnings
JSON	Full case inputs and outputs for archiving or external tools
CSV	Well pressure/temperature profile (MD, P, T, holdup)

11. Model Assumptions

The Assumptions panel in the Case Editor lists the correlations in use. Key assumptions:

- **Fluid:** Black-oil model with Standing Bo and Beggs-Robinson oil viscosity.
- **Multiphase flow:** Simplified drift-flux liquid holdup with Darcy-Weisbach friction.
- **Heat transfer:** Steady-state exponential approach to ambient (lumped UA).
- **IPR:** Linear productivity index or Vogel correlation.
- **Network:** Iterative Gauss-Seidel pressure balance for tree topology.
- **Units:** Field units throughout — psi, ft, stb/d, °F.

These are simplified engineering correlations suitable for screening studies. Results should be validated against field data or higher-fidelity simulators for critical decisions.

12. Troubleshooting

Problem	Solution
Blank page / cannot connect	Ensure both backend (port 8000) and frontend (port 3000) are running
API error when solving	Restart backend; check Terminal 1 for Python errors
No cases listed	Restart backend to trigger sample case seeding
Charts empty after Solve	Check Results panel for warnings; verify inputs are realistic
BHP less than WHP	Check rate, tubing size, and deviation survey; reduce rate or increase tubing ID
Dark/light theme	Click moon/sun icon in top-right corner

13. API Reference (Optional)

Interactive API documentation is available at <http://localhost:8000/docs> when the backend is running.

Endpoint	Method	Description
/api/health	GET	Health check
/api/cases	GET/POST	List or create cases
/api/cases/{id}	GET/PUT/DELETE	Read, update, or delete a case
/api/solve	POST	Run simulation
/api/compare	POST	Compare multiple cases
/api/sensitivity	POST	Sensitivity analysis
/api/export	POST	Export PDF, JSON, or CSV

— End of FlowSim Pro User Manual —